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Regional communities

In the 19th century, German botanists August Grisebach and Oscar Drude were among those who recognized that patterns of vegetation around the world reflect factors such as climate variations. It was clear, for example, that the typical vegetation in a wet, tropical climate was very different to that in a dry, temperate climate. Then in a landmark paper in 1899, American botanist Henry Cowles described how plants colonized sand dunes around Lake Michigan in stages—or "successions"—of increasing size and complexity.

In an influential book, *Plant Succession* (1916), Frederic Clements developed Cowles's idea, which he combined with the biogeographic thinking of the two German botanists to produce a theory of the development of natural communities.

Clements suggested that the way to understand patterns of vegetation across the world is to think in terms of "formations." A formation is a large, natural community of plants dominated by a range of life forms that reflects the regional climate. In each region, plants go through stages or successions until they reach the most complex, highly developed form of vegetation possible. Once it finally reaches this climax, the community stabilizes, in what was later termed a "steady state," and stops changing.

Clements then proposed that climax communities are bound together. Although an ecological community is made up of a multitude of plants at different stages of growth, he argued that it can be considered as a single complex organism. A community grows toward a climax in the same way that an individual develops through life stages. Clements expanded the idea to embrace all organisms in a "biome" that comprised "all the species of